(a) Not even once (b) Once
(c) Twice (d) Thrice

Ans. (b)
Perimeter of the filed
= 2 (400+300)m
=1400m
= 1.4 Km
Since A and B move in the same direction, so they will first meet each other when there is a difference of one round i.e., 1.4 Km between the two Relative speed of A and B = (3-2.5) Km
= 0.5 Km/ hr
Time take to cover 1.4 Km at the this speed:

$$=\left(\frac{1.4}{0.5}\right)\text{hr}$$

$$=2\frac{4}{5}\text{hr}$$

= 2 hr 48 min
So, they shall first cross each other at
9.48m
And again 2 hr 48 min after 9.48 am i,e.,
12.36 pm
Thus, till 12.30 pm they will cross each other once.

146. Nine different letters are to be dropped in three different letter boxes. In how many different ways can this be done?
(a) 27 (b) 3^9
(c) 93 (d) $3^9 - 3$

Ans. (b) : First letter can be dropped into any of the 3 boxes. It can be done in 3 ways. Similarly second letter can also be dropped into any of the 3 boxes in 3 ways and so on. Hence, total no of ways = 3×3×3×.....upto 9 times = 3(raise the power 9)

147. How many three-digit even numbers are there such than 9 comes as a succeeding digit in any number only when 7 is the preceding digit only when 9 is the succeeding digit?
(a) 120 (b) 210
(c) 365 (d) 405

Ans. (d) : Since, we need three digit $\overline{100^{th}}$, $\overline{10^{th}}$,& $\overline{\text{unit}}$.
There is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 are possible digits of required no. now, since required number is even no. Thus at unit place there is five possibilities - 0, 2, 4, 6 and 8.
Now at 100th place there is - 1, 2, 3, 4, 5, 6, 7, 8, 9 are possible and 10th place, 1, 2, 3, 4, 5, 6, 7, 8 and 9 are possible numbers.
Then by "fundamental theorem of counting",
9×9×5 = 405
Hence, option (d) - 405 will be correct answer.

148. In how many different ways can six players be arranged in a line such that two of them, Ajit and Mukherjee, are never together?
(a) 120 (b) 240
(c) 360 (d) 480

Ans. (d) : Total no of ways of arrangement for six players = 6! Let us take Ajit and Mukerjee as one entity. So now there are (6 – 2 + 1) = 5 players .
These 5 players can be arranged in 5! ways and Ajit and Mukerjee can be arranged among themselves in 2! ways. Thus, no of ways, when Ajit and Mukerjee are always together = 5 ! × 2!
Hence, no of ways when they are never together = Total no of ways – no of ways when they are always together
= 6 ! – (5 ! × 2 !) = 6 × 5 ! – (5 ! × 2 !) = 5 ! (6 – 2) = 480

149. 50 men or 80 women can finish a job in 50 days. A contractor deploys 40 men and 48 women for this work, but after every 'duration of 10 days, 5 men and 8 women are removed till the work is completed. The work is completed in
(a) 45 days (b) 50 days
(c) 54 days (d) 62 days

Ans. (b) : Total number of working day = 50 × 50 = 2500 days for men, For women it is equal to 4000 days For 1st 10 day means 40 men × 10 = 400 day, 48 women × 10 = 480 days 2nd after 10 days means 35 men × 10 = 350 day, 40 women × 10 = 400 days 3rd 10 day 30 men × 10 = 300 days, 32 women × 10 = 320 days 4th 10 day 25 men × 10 = 250 days, 24 women × 10 = 240 days 5th 10 day 20 men × 10 = 200 days, 16 women × 10 = 160 days Men = 1500 days, Women = 1600 days 1 man = 1.6 women 1600 days of women = 1000 day of men So, 2500 days need to complete work mean after 50 day i.e., Man of 1600 working days + woman of 1600 days i.e., 50 days.

150. Three students are picked at random from a school having a total of 1000 students. The probability that these three students will have identical date and month of their birth is
(a) $\frac{3}{1000}$ (b) $\frac{3}{365}$
(c) $\frac{1}{(365)^2}$ (d) $\frac{1}{(365)^3}$

Ans. (c) : For 1st student, Probability of selecting any one day as his birthday =365/365 =1 Now, the remaining two students to be selected must have same day as their birthday as for the 1st student. Probability of rest two students, having the same birthday as that of the 1st student = 1/365 x 1/365 Hence, required probability = 1 x 1/(365)2 = 1/(365)2

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2003

GENERAL STUDIES : PAPER-I

Time: 2 hours **Maximum Number: 200**

ANCIENT HISTORY

1. Emperor Harsha's southward march was stopped on the Narmada river by
(a) Pulakesin-I (b) Pulakesin-II
(c) Vikramaditya-I (d) Vikramaditya-II

Ans. (b) : Emperor Harsha's southward march was stopped on the Narmada river by Pulakesin II, the Chalukya king of Vatapi in northern Mysore.

2. Assertion (A) : Saluva Narasimha put an end to the old dynasty and assumed to royal life.
Reason (R) : He wanted to save the kingdom from further degeneration and disintegration.
(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (a) : The founders of Vijayanagar, Harihara and Bukka, laid the foundation of the Sangam dynasty. Virupaksha II, the last ruler of this dynasty, was a completely unfit person. There was chaos all around the empire. In this situation Narasimha Saluva, a powerful feudal lord of the empire, took power by killing Virupaksha. Narasimha Saluva strengthened the army to reorganize the empire. For that he encouraged Arab traders to import more and more horses. He appointed an ambitious man named Narsa Nayak as the guardian of his son. Narsa Nayak took over the power and greatly expanded the empire. Later, his son Veer Narasimha laid the foundation of the Tuluva dynasty.

3. Consider the following statements:
1. The Cholas defeated Pandya and Chera rulers and established their domination over peninsular India in the early medieval times.
2. The Cholas sent an expedition against Sailendra empire of South East Asia and conquered some of the areas.
Which of these statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : The Chola kingdom of the Sangam period extended from modern Tiruchi district to southern Andhra Pradesh. Their capital was first located at Uraiyur and then shifted to Puhar. Karikala was a famous king of the Sangam Cholas. Pattinappalai portrays his early life and his military conquests. In the Battle of Venni, he defeated the mighty confederacy consisting of the Cheras, Pandyas and eleven minor chieftains. The Chola emperor launched a successful naval expedition against the Sailendra kingdom.

4. 'Mrichchhakatika' and an ancient Indian book written by Shudraka deals with
(a) the love affair of a rich merchant with the daughter of a courtesan.
(b) the victory of Chandragupta II over the Shaka Kshatrapas of western India.
(c) the military expeditions and exploits of Samudragupta.
(d) the love affair between a Gupta King and a princess of Kamarupa.

Ans. (a) : Mrichchhakatika (The Clay Court) is a Sanskrit play written by Shudraka in 2nd century BC. It is about a young man named Charudatta, and his love for Vasantasena, a rich courtesan.

5. Consider the following statements:
1. Vardhaman Mahavira's mother was the daughter of Lichhavi chief Chetaka.
2. Gautama Buddha's mother was a princess from the Koshalan dynasty.
3. Parshvanatha, the twenty-third Tirthankara, belonged to Banaras.
Which of these statements is/are correct?
(a) Only 1 (b) Only 2
(c) 2 and 3 (d) 1, 2 and 3

Ans. (c) : Vardhaman Mahavira's mother, Trishala was the sister (not the daughter) of Lichhavi chief Chetaka.

6. Consider the following statements:
1. The last Mauryan ruler, Brihadratha was assassinated by his commander-in-chief, Pushyamitra Sunga.
2. The last Sunga king, Devabhuti was assassinated by his Brahmana minister Vasudeva Kanva who usurped the throne.
3. The last ruler of the Kanva dynasty was deposed by the Andhras.
Which of these statements is/are correct?
(a) 1 and 2 (b) Only 2
(c) Only 3 (d) 1, 2 and 3

Ans. (d) : All statements are correct. The last Mauryan ruler, Brihadratha was assassinated by his commander-in-chief, Pushyamitra Sunga in 185 BC. The last ruler of Kanva dynasty, Susharman was deposed by the Andhras and Satavahana dynasty came into being.